
Grounded AI Health Assistant & Emergency Triage

An Auditable Retrieval-Augmented Assistant with Rule-Based Emergency Escalation

UN SDG ALIGNMENT

Goal 3: Good Health & Well-being
Target 3.8: Access to safe, verified health information without medical misinformation.

CORE PARADIGM

Strict Negative Constraints
Refuses to guess without verified sources. Bypasses AI in medical emergencies (< 20 ms short-circuit).

PROJECT EVALUATION

Date: September 26, 2026
Status: Full Stack Deployed & Running
Tech: FastAPI · PostgreSQL (pgvector) · React · Clerk

The Critical Problem: Fluent Fabrication in Healthcare AI

1. The Danger of Fluent Fabrication

General LLMs are trained to always generate an answer. A confident, well-structured, completely fabricated paragraph about chest pain is more dangerous than an obvious typo because patients trust it blindly.

2. Total Lack of Provenance

Standard chatbots synthesize knowledge across unvetted internet forums (Reddit, WebMD, blogs). They cannot point to the legal document, paragraph, or clinical body behind a claim.

3. Emergency Neglect

General LLMs engage in conversational chit-chat ('I understand your chest hurts; have you tried resting?'). In acute crises (stroke, heart attack), every minute lost to AI generation can be fatal.

System Philosophy: Four Architectural Guardrails

Guardrail 1: Explain, Never Diagnose

The system only explains verified medical literature. It has no capability to diagnose, prescribe medications, or suggest dosages, preventing liability and clinical drift.

Guardrail 2: 100% Auditable Provenance

Every factual statement carries an inline citation chip linking directly to a specific passage in a published document from WHO, NHS, NIH, or CDC with its legal license.

Guardrail 3: Structural Refusal (0.65 Cosine Floor)

If retrieved document similarity is below 0.65, the system refuses to answer. The LLM is never called. This structurally prevents the model from hallucinating outside verified data.

Guardrail 4: Emergency Short-Circuit (< 20 ms)

Emergency symptoms (crushing chest pain, stroke, breathing failure) are caught by rule-based pre-filters on raw input, instantly routing to 112/108 without touching the LLM.

System Architecture: End-to-End Pipeline

Step 1: Auth & Rate Limit

Clerk JWT (RS256 JWS)
Redis Lua Sliding Window
10 req/min per user
(Fails open)

Step 2: Red-Flag Triage

In-Process Regex (< 1 ms)
Emergency Match -> Fixed 112/108
(No Embedding, No LLM)

Step 3: Vector Search

Gemini 768d Embeddings
Postgres 16 + pgvector HNSW
Cosine similarity floor ≥ 0.65

Step 4: Dual LLM Fallback

Primary: Groq (Llama/GPT-OSS)
Fallback: Gemini 2.5 Flash
Zero broken streams on error

Step 5: SSE Streaming UI

React 18 + Vite + Tailwind
Server-Sent Events streaming
Interactive Citation Chips

Database Design: Relational Schema & pgvector HNSW Index

documents (Corpus Metadata)

- id: UUID (Primary Key)
- title: Text (Published article title)
- source_url: Text (Original reference link)
- source_org: Text (WHO | NHS | NIH | CDC)
- license: Text (OGL-v3 | CC BY-NC-SA | Public)
- content_hash: String(64) UNIQUE (SHA-256 for idempotent re-ingestion)
- ingested_at: DateTime(timezone=True)

chunks (Vector Store)

- id: UUID (Primary Key)
- document_id: UUID (FK -> documents.id CASCADE)
- content: Text (Semantic chunk passage)
- embedding: Vector(768) (Gemini 768-dim vector)
- chunk_index: Integer (0-based chunk order)
- token_count: Integer (~300 to ~650 tokens)
- ★ INDEX: HNSW on embedding (vector_cosine_ops, m=16, ef_construction=64)

sessions (User Chat Context)

- id: UUID (Primary Key)
- clerk_user_id: Text (Indexed, user scope)
- title: Text (Auto-generated from query)
- deleted_at: DateTime (Nullable, soft delete)
- created_at: DateTime(timezone=True)

messages (Audit Trail & Tracking)

- id: UUID (Primary Key)
- session_id: UUID (FK -> sessions.id CASCADE)
- role: Text ('user' | 'assistant')
- content: Text (Message body or response)
- retrieved_chunk_ids: UUID[] (Audit trail)
- llm_provider: Text ('groq' | 'gemini')
- latency_ms: Integer | was_red_flag: Boolean

API Architecture: REST & Server-Sent Events (SSE) Endpoints

CORE API SPECIFICATION

- **POST /api/chat/stream [Bearer <JWT>]**
Payload: {session_id, content} → SSE stream with event types: token, citation, escalation, done
- **GET /api/sessions [Bearer <JWT>]**
Payload: None → Returns 200 OK with list of active user chat sessions [SessionOut]
- **GET /api/sessions/{id}/messages [Bearer <JWT>]**
Payload: session_id → Returns 200 OK with full conversation history & citation IDs [MessageOut]
- **DELETE /api/sessions/{id} [Bearer <JWT>]**
Payload: session_id → Soft-deletes session (sets deleted_at); returns 204 No Content
- **GET /api/me [Bearer <JWT>]**
Payload: None → Returns 200 OK with authenticated Clerk user ID & email (401 if missing/invalid)
- **GET /health [Public]**
Payload: None → Returns 200 OK: {status, database: {reachable, pgvector}, redis: {reachable}}

Security & Middleware Layer

- RequestContextMiddleware: Injects unique X-Request-ID across all logs.
- Clerk JWKS Authenticator: Validates RS256 JWT tokens cryptographically.
- Redis Sliding Window: Enforces 10 req/min rate limit per user.

SSE Streaming Protocol (ReadableStream)

- Uses fetch + ReadableStream instead of native EventSource to support Authorization headers.
- Dispatches real-time structured JSON chunks to React frontend.
- Automatically cleans up DB sessions and logs response latency.

The Medical Reference Corpus: Quality Over Unvetted Breadth

Curated & License-Checked Repository

- 42+ Vetted Source Documents: Covering infectious diseases (dengue, malaria, TB), chronic conditions (diabetes, hypertension, asthma), everyday symptoms, and prevention.
- Strict Licensing Compliance: Every source explicitly mapped in corpus_manifest.json:
 - WHO: CC BY-NC-SA 3.0 IGO
 - UK NHS: Open Government Licence (OGL v3.0)
 - US NIH / CDC: US Federal Public Domain
- Deterministic Extraction: Handled via PyMuPDF and Trafilatura, stripping boilerplate, headers, and footers.
- Semantic Chunking: 153 chunks bounded between ~300 to ~650 tokens to ensure clinical coherence.

High Priority for Indian Healthcare

- High-Relevance Infectious Disease Focus: Extensive coverage for Dengue, Malaria, Tuberculosis, Typhoid, and Diarrhoeal diseases.
- Everyday Family Symptoms: High fever in adults & children, cough, dehydration, vomiting, paediatric skin rashes.
- Deliberate Frozen Corpus: Frozen snapshot prevents silent data corruption or unvetted web drift. If a condition isn't verified, it safely refuses.
- Auditable Citations: Every answer references the exact source document title, URL, and medical authority.

Safety Engineering: Deterministic Emergency Short-Circuit

PERFORMANCE ASYMMETRY: EMERGENCY TRIAGE vs GENERATION

Red-Flag Matcher: < 1 ms in-process (~20 ms HTTP) vs LLM Generation: ~2,512 ms

Short-circuits immediately before embedding, retrieval, and LLM generation. Eliminates latency and non-deterministic risks.

9 Critical Emergency Categories

1. Cardiac (crushing chest pain, arm radiation)
2. Breathing Distress (choking, severe asthma)
3. Stroke Signs (BE-FAST: facial droop, arm weakness)
4. Uncontrolled Severe Bleeding
5. Acute Head Injury with altered mental state
6. Loss of Consciousness / Unresponsiveness
7. Anaphylaxis (throat swelling, severe allergy)
8. Seizures / Convulsions
9. Mental Health Crisis / Self-Harm

Emergency Helplines & Asymmetric Cost

Integrated Indian Emergency Services:

- 112: All-in-One National Emergency Helpline
- 108: Emergency Ambulance Response
- 14416: Tele-MANAS (Mental Health Helpline)

The 'Cost Asymmetry' Rule:

If a user says 'I know it's just acid reflux, but my chest feels tight', the system still escalates. An unnecessary emergency checkup is an annoyance; a missed heart attack is fatal.

High Availability & Operational Resilience

Dual-Provider Automatic LLM Fallback

- Primary Provider: Groq (Llama / GPT-OSS 120B) for ultra-fast token streaming (~1,800 ms median latency).
- Automatic Fallback: Gemini 2.5 Flash triggers automatically if Groq hits rate limits (TPM) or connection timeouts (8s limit).
- Clean Stream Recovery: If the primary fails mid-generation, the client cleanly discards the partial buffer and resumes from the fallback. No broken hybrid text.
- Measured Fallback Rate: 44% under heavy testing due to free-tier token budgets, proving the fallback is actively exercised.

Redis Sliding-Window Rate Limiting

- Per-User Sliding Window: 10 requests per minute enforced per authenticated Clerk User ID.
- Atomic Lua Script: Trims old timestamps, counts active requests, records timestamp, and sets TTL in a single atomic Redis round-trip.
- Eliminates Race Conditions: Traditional read-then-write approaches permit concurrent limit bypass. Atomic execution prevents this.
- Fail-Open Resiliency: If Redis is unavailable, rate limiting is bypassed. A rate-limiter failure never takes down critical health access.

Current Implementation Status: Production Ready

Backend & Database

FastAPI (Python 3.11, async)
SQLAlchemy 2.0 + asyncpg
PostgreSQL 16 + pgvector (HNSW)
Redis 7 Ephemeral Cache
Alembic Database Migrations

Frontend & UX

React 18 + TypeScript + Vite
Tailwind CSS modern styling
Server-Sent Events (SSE) Reader
Interactive Citation Inspector
Emergency Visual Escalation

Security & Auth

Clerk Authentication
JWKS Cryptographic Verification
RS256 JWT validation on all API calls
No browser-side credential trust
Strict CORS & rate limit headers

RIGOROUS AUTOMATED TEST SUITE: 282 UNIT TESTS (100% PASSING)

Comprehensive test coverage executed in 4.46 seconds:

- 98 tests for red-flag emergency detection & phrasing variants
- 38 tests for vector retriever & floor filtering
- 31 tests for document extraction & chunking bounds
- 21 tests for authentication & rate limiting

Empirical Benchmark Results: Grounding & Latency

Retrieval & Grounding Precision

- In-Corpus Retrieval Hit Rate: 90% (27/30 in-corpus medical queries retrieved relevant passages).
- Out-of-Corpus False Hit Rate: 0.0% (0/8 non-medical or off-topic queries cleared the 0.65 threshold).
- Top Similarity Separation: In-corpus median: 0.727 vs Out-of-corpus max: 0.555 (Wide 0.172 margin).
- Citation Validity: 100% (129 of 129 generated citations mechanically verified to match retrieved source text).
- Mean Citations: 4.8 verified sources cited per response.

System Latency Profile

- Red-Flag Escalation: ~0.1 ms in-process matcher (HTTP round-trip ~20 ms).
- Query Embedding (Gemini): Median 714 ms (p95: 950 ms).
- Vector Retrieval (pgvector HNSW): Median 46 ms (p95: 48 ms).
- Time to First Token (TTFT): Median 1,129 ms.
- Total End-to-End Generation: Median 2,512 ms (Groq p50: 1,843 ms, Gemini fallback p50: 3,605 ms).
- Safety Short-Circuit: 100% of red-flag queries prevented from reaching the LLM.

Live Demonstration: Four Key Evaluation Beats

Beat 1: Grounded Answer

Query: 'What causes iron deficiency anaemia?'

Shows: Real-time SSE streaming, inline citation chips, and source inspector panel displaying NHS/WHO origin.

Beat 2: Grounding Proof

Query: 'What is the capital of France?'

Shows: Immediate refusal. Proves the model is structurally blocked without verified medical context.

Beat 3: Emergency Short-Circuit

Query: 'Crushing chest pain radiating to arm'

Shows: Instant red emergency banner (~20 ms). Directs to 112/108 without waiting for LLM tokens.

Beat 4: Cost Asymmetry

Query: 'I think it's acid reflux but chest hurts'

Shows: Still escalates. Proves safety priority: false alarms are minor; missed heart attacks are fatal.

Defense: Why Use This Over Frontier AI Models (ChatGPT/Gemini)?

Frontier AI Models (ChatGPT, Claude)

- Unconstrained Intelligence: Trained to be helpful and conversational. They will speculate on symptoms and suggest unverified home remedies.
- Opaque Sourcing: Synthesize text from internet scrapings (Reddit, forums, blogs). Zero legal traceability or document-level citations.
- Dangerous Latency in Crises: Take 2-5 seconds over HTTP to chat conversationally when a user is experiencing cardiac arrest or stroke.
- Prompt Drift & Jailbreaks: System prompts ('Act like a doctor') can be bypassed or ignored by creative user queries.

Our Grounded Assistant Architecture

- Enforced Negative Constraints: Architecturally incapable of hallucinating without verified medical context (0.65 similarity floor).
- 100% Chain-of-Custody: Every claim links to an exact legal document chunk from WHO, NHS, NIH, or CDC.
- Deterministic Emergency Speed: Rule-based regex evaluates raw message in < 1 ms (~20 ms HTTP) and routes to 112/108 immediately.
- Zero Diagnosis Liability: Strictly explains literature; cannot prescribe medication or suggest dosages by design.

Engineering Boundaries & Future Roadmap

Current Engineering Boundaries

- Curated Frozen Corpus: Limited to 42-50 vetted documents. Questions outside this scope (e.g. rare diseases, COVID-19) are refused.
- Informational Only: Built strictly as an informational tool. Cannot replace physical medical exams, diagnostics, or clinical triage.
- English Language: Currently trained and indexed for English medical literature.
- Manual Page Extraction for Bot-Protected Sites: 8 CDC articles require manual saving due to strict anti-scraping Cloudflare blocks.

Future Technical Roadmap

- Automated LLM-as-a-Judge Verification: Adding real-time claim-to-passage entailment verification (NLI) before streaming.
- Indian Regional Language Support: Expanding ingestion and queries to Hindi, Tamil, Telugu, and Bengali.
- National Health Authority Ingestion: Expanding corpus to include official publications from ICMR (Indian Council of Medical Research) and AIIMS.
- Offline Edge Deployment: Exploring quantized on-device embeddings and small language models for remote rural health clinics.

Summary: An Assistant You Can Audit and Trust

Safety-Critical AI requires architectural constraints, deterministic triage, and legal provenance.

100%

Grounded Citation Validity (Verified against retrieved chunks)

< 20 ms

Emergency Triage Short-Circuit to 112 / 108

282

Passing Automated Unit Tests covering all safety layers